**Q1 JEE Main 2020 - 7 Jan (Morning)**

The logical statement

$(p \rightarrow q) \wedge (q \rightarrow \sim p)$ is equivalent to

(A) $\sim p$
(B) $p$
(C) $p \wedge q$
(D) $p \vee q$

**Q2 JEE Main 2020 - 7 Jan (Evening)**

Let $A,\ B,\ C$ and $D$ be four non-empty sets. The contra-positive statement of " If $A \subseteq B$ and $B \subseteq D$, then $A \subseteq C$" is:

(A) If $A \nsubseteq C$ then $A \subseteq B$ and $B \subseteq D$
(B) If $A \nsubseteq C$ then $A \nsubseteq B$ and $B \subseteq D$
(C) If $A \subseteq C$ then B $\subset A$ and $D \subset B$
(D) If $A \nsubseteq C$ then $A \nsubseteq B$ and $B \nsubseteq D$

**Q3 JEE Main 2020 - 8 Jan (Morning)**

Which one of the following is a tautology?

(A) $(p \wedge (p \rightarrow q)) \rightarrow q$
(B) $q \rightarrow p \wedge (p \rightarrow q)$
(C) $p \vee (p \wedge q)$
(D) $p \wedge (p \vee q)$

**Q4 JEE Main 2020 - 8 Jan (Evening)**

Which of the following statement is a tautology ?

(A) $\sim (p \vee \sim q) \rightarrow p \ \vee q$
(B) $p \ \vee (\sim q) \rightarrow p \vee q$
(C) $\sim (p \wedge \sim q) \rightarrow p \vee q$
(D) $\sim (p \vee \sim q) \rightarrow p \wedge q$

**Q5 JEE Main 2020 - 9 Jan (Morning)**

The negation of '$\sqrt{5}$ is an integer or $5$ is an irrational number' is

(A) $\sqrt{5}$ is an integer and $5$ is not an irrational Number

(B) $\sqrt{5}$ is not an integer and $5$ is an irrational Number
(C) $\sqrt{5}$ is not an integer or $5$ is not an irrational Number
(D) $\sqrt{5}$ is not an integer and $5$ is not an irrational Number

**Q6 JEE Main 2020 - 9 Jan (Evening)**

If $p \rightarrow (p \wedge \sim q)$ false, then the truth values of $p$ and $q$ are respectively :

(A) $F,\ F$
(B) $F,\ T$
(C) $T,\ F$
(D) $T,\ T$

**Q7 JEE Main 2020 - 2 Sep (Morning)**

The contrapositive of the statement "If I reach the station in time, then I will catch the train" is:

(A) If $\mathrm{I}$ will catch the train, then $\mathrm{I}$ reach the station in time
(B) If 1 do not reach the station in time, then 1 will catch the train
(C) If 1 do not reach the station in time, then 1 will not catch the train
(D) If $\mathrm{I}$ will not catch the train, then $\mathrm{I}$ do not reach the station in time

**Q8 JEE Main 2020 - 2 Sep (Evening)**

Which of the following is a tautology?

(A) $(\sim p) \wedge (p \vee q) \rightarrow q$
(B) $(\sim \mathrm{q}) \vee (\mathrm{p} \wedge \mathrm{q}) \rightarrow \mathrm{q}$
(C) $(p \rightarrow q) \wedge (q \rightarrow p)$
(D) $(q \rightarrow p) \vee \sim (p \rightarrow q)$

**Q9 JEE Main 2020 - 3 Sep (Morning)**

The proposition $p \rightarrow \sim (p \wedge \sim q)$ is equivalent to:

(A) q
(B) $(\sim p) \wedge q$
(C) $(\sim \mathrm{p}) \vee (\sim \mathrm{q})$
(D) $(\sim p) \vee q$

**Q10 JEE Main 2020 - 3 Sep (Evening)**

Let $p, q, r$ be three statements such that the truth value of $(\mathrm{p} \wedge \mathrm{q}) \rightarrow (\sim \mathrm{q} \vee \mathrm{r})$ is $\mathrm{F}$. Then the truth values of $\mathrm{p}, \mathrm{q}, \mathrm{r}$ are respectively :

(A) $\mathrm{T}, \mathrm{F}, \mathrm{T}$
(B) $\mathrm{F}, \mathrm{T}, \mathrm{F}$
(C) $T, T, T$
(D) $\mathrm{T}, \mathrm{T}, \mathrm{F}$

**Q11 JEE Main 2020 - 4 Sep (Morning)**

Given the following two statements:
$(s_1) : (q \vee p) \rightarrow (p \leftrightarrow \sim q)$ is a tautology :
$(s_2) : \sim q \wedge (\sim p \leftrightarrow q)$ is a fallacy. Then :

(A) only $(\mathrm{S}_1)$ is correct.
(B) both $(s_1)$ and $(s_2)$ are correct.
(C) only $(\mathrm{S}_2)$ is correct.
(D) both $(S_1)$ and $(S_2)$ are not correct.

**Q12 JEE Main 2020 - 4 Sep (Evening)**

Contrapositive of the statement:
'If a function $f$ is differentiable at a, then it is also continuous at a', is :

(A) If a function $f$ is continuous at a, then it is differentiable at a.
(B) If a function $f$ is not continuous at a, then it is not differentiable at a.
(C) If a function $f$ is not continuous at a, then it is differentiable at a.
(D) If a function $f$ is continuous at a, then it is not differentiable at a.

**Q13 JEE Main 2020 - 5 Sep (Morning)**

The negation of the Boolean expression $x \leftrightarrow \sim y$ is equivalent to :

(A) $(x \wedge \sim y) \vee (\sim x \wedge y)$
(B) $(x \wedge y) \vee (\sim x \wedge \sim y)$
(C) $(x \wedge y) \wedge (\sim x \vee \sim y)$
(D) $(\sim x \wedge y) \vee (\sim x \wedge \sim y)$

**Q14 JEE Main 2020 - 5 Sep (Evening)**

The statement $(\mathrm{p} \rightarrow (\mathrm{q} \rightarrow \mathrm{p})) \rightarrow (\mathrm{p} \rightarrow (\mathrm{p} \vee \mathrm{q}))$ is

(A) a tautology

(B) equivalent to $(\mathrm{p} \vee \mathrm{q}) \wedge(\sim \mathrm{p})$
(C) a contradiction
(D) equivalent to $(p \wedge q) \vee(\sim q)$

**Q15 JEE Main 2020 - 6 Sep (Morning)**

The negation of the Boolean expression $p \vee(\sim p \wedge q)$ is equivalent to :

(A) $\sim p \vee q$
(B) $p \wedge \sim q$
(C) $\sim p \vee \sim q$
(D) $\sim p \wedge \sim q$

**Q16 JEE Main 2020 - 6 Sep (Evening)**

Consider the statement: "For an integer n, if $n^{3}-1$ is even, then $n$ is odd." The contrapositive statement of this statement is :

(A) For an integer $n$, if $n$ is odd, then $n^{3}-1$ is even
(B) For an integer $n$, if $n$ is even, then $n^{3}-1$ is even.
(C) For an integer $n$, if $n$ is even, then $n^{3}-1$ is odd.
(D) For an integer $n$, if $n^{3}-1$ is not even, then $n$ is not odd.

# Answer Key

| | | | |
|---|---|---|---|
| **Q1** (A) | **Q2** (D) | **Q3** (A) | **Q4** (A) |
| **Q5** (D) | **Q6** (D) | **Q7** (D) | **Q8** (A) |
| **Q9** (D) | **Q10** (D) | **Q11** (D) | **Q12** (B) |
| **Q13** (B) | **Q14** (A) | **Q15** (D) | **Q16** (C) |

**Q1**

| $p$ | $q$ | $p \to q$ | $\sim p$ | $q \to \sim p$ | $(p \to q) \wedge (p \to \sim q)$ |
|---|---|---|---|---|---|
| $T$ | $T$ | $T$ | $F$ | $F$ | $F$ |
| $T$ | $F$ | $F$ | $F$ | $T$ | $F$ |
| $F$ | $T$ | $T$ | $T$ | $T$ | $T$ |
| $F$ | $F$ | $T$ | $T$ | $T$ | $T$ |

Clearly $(p \to q) \wedge (q \to \sim p)$ is equivalent to $\sim p$

**Q2**

Let $P = A \subset B,\ Q = B \subset C,\ R = C \subset A$

Contra-positive of $(P \wedge Q) \to R$ is $\sim R \to \sim (P \wedge Q) = \sim\ R \to \sim\ (P) \vee \sim\ (Q)$

**Q3**

| p | q | p→ q | p∧ (p → q) | (p ∧(p →q)) → q | q→(p∧(p→q)) |
|---|---|---|---|---|---|
| T | T | T | T | T | T |
| T | F | F | F | T | T |
| F | T | T | F | T | F |
| F | F | T | F | T | T |

| p | q | p→ q | p ∧ q | p ∨ (p ∧q ) | p ∨ q | p∧ (p∨ q) |
|---|---|---|---|---|---|---|
| T | T | T | T | T | T | T |
| T | F | F | F | T | T | T |
| F | T | T | F | F | T | F |
| F | F | T | F | F | F | F |

FALSE

**Q4**

$(\sim p \wedge q) \to (p \vee q)$

$\sim \{(\sim p \wedge q) \wedge (\sim p \wedge \sim q)\}$

$\sim \{p \wedge f\}$

**Q5**

$\sqrt{5}$ is not an integer and 5 is not an irrational Number $\sim (p \vee q) = \sim p \wedge \sim q$

**Q6**

| $p$ | $q$ | $\sim q$ | $p \wedge \sim q$ | $p \to (p \wedge \sim q)$ |
|---|---|---|---|---|
| $T$ | $T$ | $F$ | $F$ | $F$ |
| $T$ | $F$ | $T$ | $T$ | $T$ |
| $F$ | $T$ | $F$ | $F$ | $T$ |
| $F$ | $F$ | $T$ | $F$ | $T$ |

**Q7**

Contrapositive of if $P$ then $Q$ is

if not $Q$ then not $P$

$\therefore$ If I will not catch the train then I do not reach the station in time.

**Q8**

Truth table

\begin{tabular}{|

$\sim p \wedge (p \vee q) \rightarrow q$ be a tautology

**Q9**

$p \rightarrow \sim (p \wedge \sim q)$ is equivalent to $\sim$ pvq

**Q10**

$(p \wedge q) \rightarrow (\sim q \vee r)$
$= \sim (\mathbf{p} \wedge \mathbf{q}) \vee (\sim \mathbf{q} \vee \mathbf{r})$
$= (\sim p \vee \sim q) \vee (\sim q \vee r)$
$= (\sim p \vee \sim q \vee r)$
$\because \sim \mathrm{p} \vee \sim \mathrm{q} \vee \mathrm{r}$ is false, then
$\sim p, \sim q$ and $r$ all three must be false.
$\Rightarrow$ $\mathrm{p}$ is true, $\mathrm{q}$ is true and $\mathrm{r}$ is false.

**Q11**

| p | q | ~p | ~q | q∨p | p↔~q | $(S_1)$ | ~p↔q | $(S_2)$ |
|---|---|---|---|---|---|---|---|---|
| T | T | F | F | T | F | F | F | F |
| T | F | F | T | T | T | T | T | T |
| F | T | T | F | T | T | T | T | F |
| F | F | T | T | F | F | T | F | F |

$(S_1)$: not tautology
$(S_2)$: not fallacy

$\therefore$ Both $(\mathrm{S}_1)$ and $(\mathrm{S}_2)$ are incorrect.

**Q12**

Contrapositive statement will be
If a function is not continuous at 'a', then it is not differentiable at 'a'.

**Q13**

$\mathbf{p} : \mathbf{x} \leftrightarrow \sim \mathbf{y} = (\mathbf{x} \rightarrow \sim \mathbf{y}) \wedge (\sim \mathbf{y} \rightarrow \mathbf{x})$
$= (\sim \mathbf{x} \vee \sim \mathbf{y}) \wedge (\mathbf{y} \vee \mathbf{x})$
$= \sim (\mathbf{x} \wedge \mathbf{y}) \wedge (\mathbf{x} \vee \mathbf{y})$
Negation of $\mathbf{p}$ is $\sim \mathbf{p} = (\mathbf{x} \wedge \mathbf{y}) \vee \sim (\mathbf{x} \vee \mathbf{y})$
$= (\mathbf{x} \wedge \mathbf{y}) \vee (\sim \mathbf{x} \wedge \sim \mathbf{y})$

**Q14**

$(p \rightarrow (q \rightarrow p)) \rightarrow (p \rightarrow (p \vee q))$ is
By truth table

\begin{tabular}{|c|c|c|c|c|c|c|} \hline \( \mathrm{p} \) & \( \mathrm{q} \) & \( \mathrm{p} \mathrm{v} \mathrm{q} \) & \( \mathrm{p} \righ

So it is tautology.

**Q15**

$\sim(\mathbf{p} \vee(\sim \mathbf{p} \wedge \mathbf{q}))$
$=\sim \mathbf{p} \wedge(\mathbf{p} \vee \sim \mathbf{q})$
$=(\sim \mathbf{p} \wedge \mathbf{p}) \vee(\sim \mathbf{p} \wedge \sim \mathbf{q})$
$=\sim \mathbf{p} \wedge \sim \mathbf{q}$

**Q16**

Contrapositive statement will be
"For an integer $n,$ if $n$ is not odd then $n^{3}-1$ is not even".
OR
"For an integer $n,$ if $n$ is even then $n^{3}-1$ is odd.